Lecture Script
Attention and Transformers

Outline

1. Data-independent weights

An MLP layer with $x \in \mathbb{R}^d$, $W \in \mathbb{R}^{m\times d}$:

$$y = \sigma(Wx + b)$$

W is **fixed after training**. Every input gets multiplied by the same W—**data-independent** weights.

What does W actually do?

Row i of W computes the i-th output component: $[Wx]_i = w_i^\top x$, where $w_i \in \mathbb{R}^d$. A fixed dot product—a fixed linear measurement of x. The input has no say in *what gets measured*.

Sets of vectors

Now suppose we have n vectors $x_1, \ldots, x_n \in \mathbb{R}^d$ and want a representation y_i for each. In Einstein notation, the natural generalization of $y_i = W_{ij} x_j$ is:

$$y_{ik} = W_{ij}\, x_{jk}$$

This *does* mix across elements—each y_i is a linear combination of all x_j, applied independently per feature k. But:

- $W \in \mathbb{R}^{n \times n}$ depends on the set size. Breaks for variable n.
- The mixing weights are **fixed**—element 3 always gets the same weight from element 7, regardless of what they contain.

For a single vector (n scalars), fixed mixing is fine—stack layers with nonlinearities and you can learn any mapping. For a set of vectors, the *relationships between elements* are themselves data-dependent: which elements should inform which depends on their content. Fixed weights can learn this given enough depth, but it's a poor inductive bias—like using MLPs on images instead of convolutions.

We want $y_i = f(x_i, \{x_j\}_{j=1}^n)$ with **data-dependent** mixing that works for any n:

$$y_i = \sum_{j=1}^{n} \alpha_{ij}\, x_j$$

where α_{ij} are not learned parameters—they're computed from the inputs. We'll build up α_{ij} in stages.

2. Building up attention

Step 1: Raw dot-product similarity (nothing learned)

Weight by similarity. Similarity = dot product, softmax to normalize:

$$\alpha_{ij} = \frac{\exp(x_i^\top x_j)}{\sum_{k=1}^n \exp(x_i^\top x_k)}$$

$\alpha_{ij} \geq 0$, $\sum_j \alpha_{ij} = 1$. Element i pulls more from elements similar to it. No learned parameters.

In matrix form, stacking inputs as rows of $X \in \mathbb{R}^{n\times d}$:

$$Y = \underbrace{\text{softmax}(XX^\top)}_{\text{weights} \in \mathbb{R}^{n\times n}} \underbrace{X}_{\text{values to mix}}$$

$XX^\top \in \mathbb{R}^{n\times n}$ is pairwise similarity; softmax normalizes each row. Compare to the fixed mixing $Y = WX$ from before: same structural role ($n \times n$ matrix left-multiplying X), but now the mixing weights are computed from the input rather than learned.

Step 2: Learned metric (one matrix)

Project before computing the score. With a learned $W_s \in \mathbb{R}^{d'\times d}$:

$$s_{ij} = (W_s x_i)^\top (W_s x_j) = x_i^\top \underbrace{W_s^\top W_s}_{\text{learned metric}} x_j$$

Then $\alpha_{ij} = \text{softmax}_j(s_{ij})$ and $y_i = \sum_j \alpha_{ij} x_j$ as before. This learns a Mahalanobis-like metric—reshapes what "similar" means. But query and key are still the same function of x, and we still return raw x_j.

Step 3: Separate query and key (two matrices)

$$s_{ij} = (W_Q x_i)^\top (W_K x_j)$$

Again $\alpha_{ij} = \text{softmax}_j(s_{ij})$, $y_i = \sum_j \alpha_{ij} x_j$. Now "what am I looking for?" ($W_Q x_i$) is decoupled from "what do I advertise?" ($W_K x_j$). The matching is asymmetric. But still returning raw x_j.

Step 4: Separate values (three matrices — full attention)

$$y_i = \sum_j \text{softmax}_j\left(\frac{(W_Q x_i)^\top (W_K x_j)}{\sqrt{d_k}}\right) W_V x_j$$

Three decoupled roles:

- **Query** $q_i = W_Q x_i \in \mathbb{R}^{d_k}$: what element i is looking for
- **Key** $k_j = W_K x_j \in \mathbb{R}^{d_k}$: what element j advertises
- **Value** $v_j = W_V x_j \in \mathbb{R}^{d_v}$: what element j hands over when selected (now $y_i \in \mathbb{R}^{d_v}$, not $\mathbb{R}^d$)

In matrix form (projections multiply on the right since elements are rows of X):

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$

The $\sqrt{d_k}$: dot products grow with dimension, pushing softmax toward one-hot. Dividing by $\sqrt{d_k}$ keeps gradients stable.

The progression: raw dot product → learned metric → asymmetric matching → asymmetric matching with decoupled retrieval. Each step adds a degree of freedom in how elements talk to each other.

$A = \text{softmax}(QK^\top/\sqrt{d_k}) \in \mathbb{R}^{n\times n}$. Row i is a distribution over elements—the attention pattern for i. Not symmetric.

Permutation equivariant: permute rows of X, rows/columns of $QK^\top$ permute consistently, output permutes the same way. No assumption of sequence or order.

3. The transformer block

What attention can't do

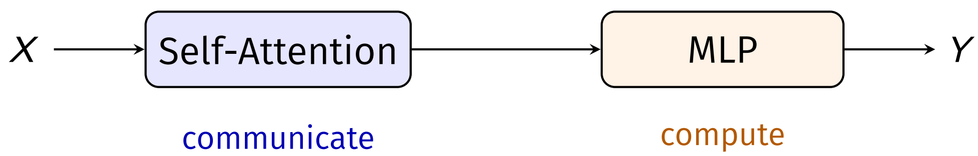
$y_i = \sum_j \alpha_{ij} v_j$ is a **weighted average** of value vectors. Attention is nonlinear overall (the softmax makes α_{ij} a nonlinear function of the input), but *for a given set of weights*, y_i is just a convex combination of $\{v_j\}$. There's no elementwise nonlinearity applied to the result—no capacity to transform what was gathered.

Adding an MLP

Follow attention with an MLP applied independently per element. (In practice $d_v = d$, so attention preserves dimension.) With $W_1 \in \mathbb{R}^{d_{\text{ff}} \times d}$, $W_2 \in \mathbb{R}^{d \times d_{\text{ff}}}$:

$$\text{MLP}(y_i) = W_2\,\sigma(W_1 y_i + b_1) + b_2$$

The **transformer block**:



Attention = *inter*-element (communicate). MLP = *intra*-element (compute).

4. The residual stream

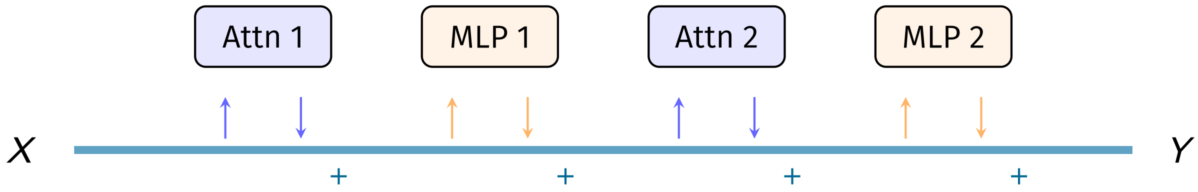
Stacking blocks without skip connections makes it hard to preserve earlier representations, and gradients tend to vanish through many layers.

Residual connections (from ResNets): $y = x + f(x)$. The layer learns the *change*, not the whole representation. Applied around both sub-layers:

$$X' = X + \text{Attention}(X)$$
$$X'' = X' + \text{MLP}(X')$$

(Also: layer normalization before each sub-layer, the "pre-norm" convention. Stabilizes training.)

Residual stream: each layer reads from and writes back to a shared representation.



Information accumulates—layer 5 can still read what layer 1 wrote, or the original input embedding.

5. Multi-head attention

The limitation of a single head

One head = one $n \times n$ attention matrix. A single pattern has to compromise between different reasons to attend.

Multiple heads in parallel

Run h **attention heads in parallel**, each with its own projections in dimension $d_k = d/h$. For head $\ell = 1, \ldots, h$:

$$Q^{(\ell)} = XW_Q^{(\ell)}, \quad K^{(\ell)} = XW_K^{(\ell)}, \quad V^{(\ell)} = XW_V^{(\ell)}$$

Concatenate and project back:

$$\boxed{\mathrm{MultiHead}(X) = \mathrm{Concat}(\mathrm{head}_1, \ldots, \mathrm{head}_h)\, W_O}$$

where $\mathrm{head}_\ell = \mathrm{Attention}(Q^{(\ell)}, K^{(\ell)}, V^{(\ell)}) \in \mathbb{R}^{n \times d_k}$ and $W_O \in \mathbb{R}^{d \times d}$.

Each head operates in $\mathbb{R}^{d/h}$. Total parameters: same as one full-dimension head—a **repartitioning**, not an expansion. Payoff: h different $n \times n$ attention matrices.

6. Positional information

Everything so far is permutation equivariant. For **sequences**, order matters: "the cat sat on the mat" ≠ "the mat sat on the cat."

Add positional information to the input embeddings before the transformer.

Given input embeddings $x_1, \ldots, x_n$ and position encodings $p_1, \ldots, p_n$:

$$\tilde{x}_i = x_i + p_i$$

Permuting the inputs now changes the representation—symmetry broken.

Sinusoidal (original). The original transformer used fixed sinusoidal functions at different frequencies:

$$p_{i,2k} = \sin(i/10000^{2k/d}), \qquad p_{i,2k+1} = \cos(i/10000^{2k/d})$$

Each dimension oscillates at a different frequency—coarse to fine position. No learned parameters.

Learned. $p_1, \ldots, p_{n_{\max}}$ as a lookup table of learned vectors. Can't extrapolate beyond $n_{\max}$.

Positional encoding is a **modular choice**: sets don't need it, sequences do. The transformer is a set-processing architecture by default.